



28th Linux SIG Workshop

Making mkCDrec PXE aware

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mkCDrec – Make CD-ROM Recovery

- ◆ El Torito bootable CD-ROM based on Linux/x86
- ◆ Disaster Recovery tool-box
- ◆ Usability of mkCDrec
 - ◆ Recovery tool (rescue mode)
 - ◆ Restore complete system from CD/DVD/tape/NFS
 - ◆ Clone a new system
 - ◆ Restore single file system (FS type may change)
- ◆ Open Source!!! See <http://mkcdrec.ota.be/>

mkCDrec - History

- ◆ Coding started begin November 2000
- ◆ Official release 25 November 2000 (v0.3)
- ◆ Booting is possible via
 - ◆ Floppy (using syslinux)
 - ◆ CD-ROM (using syslinux or isolinux)
 - ◆ Network (using pxelinux)
 - ◆ Tape (not yet implemented - OBDR)

mkCDrec Utilities

- ◆ Parted – resize partitions
- ◆ Partition Image – a ghost clone
- ◆ Memtest86 – memory check tool
- ◆ Chntpw – change NT admin passwd
- ◆ Cfg2html – system configuration report
- ◆ Smartctl – report disk health condition
- ◆ Clam Antivirus – virus checker

mkCDrec - Download

- ◆ Download the mkCDrec sources or RPM
 - ◆ RPM is precompiled and installs in /var/opt/mkcdrec
 - ◆ Sources need a gcc compiler and other tools
 - ◆ Make test: to check missing binaries
- ◆ Download mkCDrec Utilities afterwards
 - ◆ Install under mkCDrec top directory
- ◆ Pre-build images to burn on CDR
- ◆ mkCDrec Howto

mkCDrec - Usage

- ◆ Make test
 - ◆ Lists the missing software (needed by mkcdrec)
 - ◆ Edit Config.sh file if needed, e.g. SCSIDEVICE
- ◆ Make rescue
 - ◆ Build an ISO9660 image without backups
- ◆ Make CD-ROM
 - ◆ Build an ISO9660 image with backups on CD/DVD
- ◆ Make help

```
gdha@localhost: /var/opt/mkcdrec - Shell - Konsole

Session Edit View Bookmarks Settings Help

Test 6:  romfs supported  by the kernel?      N/A
Test 7:  cramfs supported  by the kernel?      N/A
Test 8:  strip (from binutils) available?      Passed
Test 9:  BOOT_FLOPPY_DENSITY=ED ok?            Passed
Test 10: cdrecord -scanbus                     Passed
Warning: Set your SCSIDEVICE to 0,0,0 in Config.sh
Test 11: Header files present?                 Passed
Test 12: DEVFS supported by kernel?            FAILED
Warning: Edit Config.sh and make DEVFS=1
Test 13: filesystem tools present?
        ext2:                                  Passed
        ext3:                                  Passed
Test 14: initrd must be compiled in kernel!    Passed
Test 15: Amount of memory available            261 Mb
Test 16: scripts/Config.sh a link?            Warning
        Will correct it immediately!
Test 17: serial console                       N/A
Test 18: supported architecture?              Passed

*****
*      Oops! I encountered 1 problem(s)!      *
*      Please correct it, or install the missing components.  *
*****
make[1]: Leaving directory `/var/opt/mkcdrec'
[root@dhcp9 mkcdrec]#
```

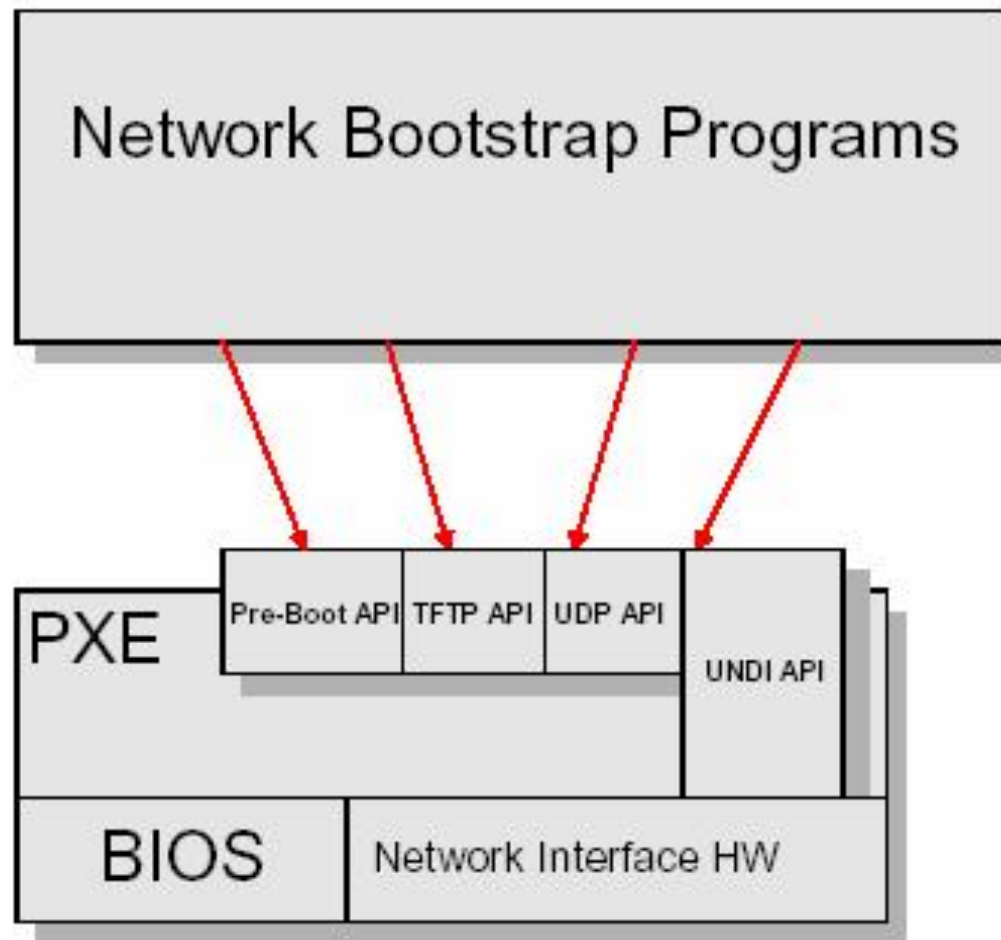
PXE: Pre-boot Execution Environment

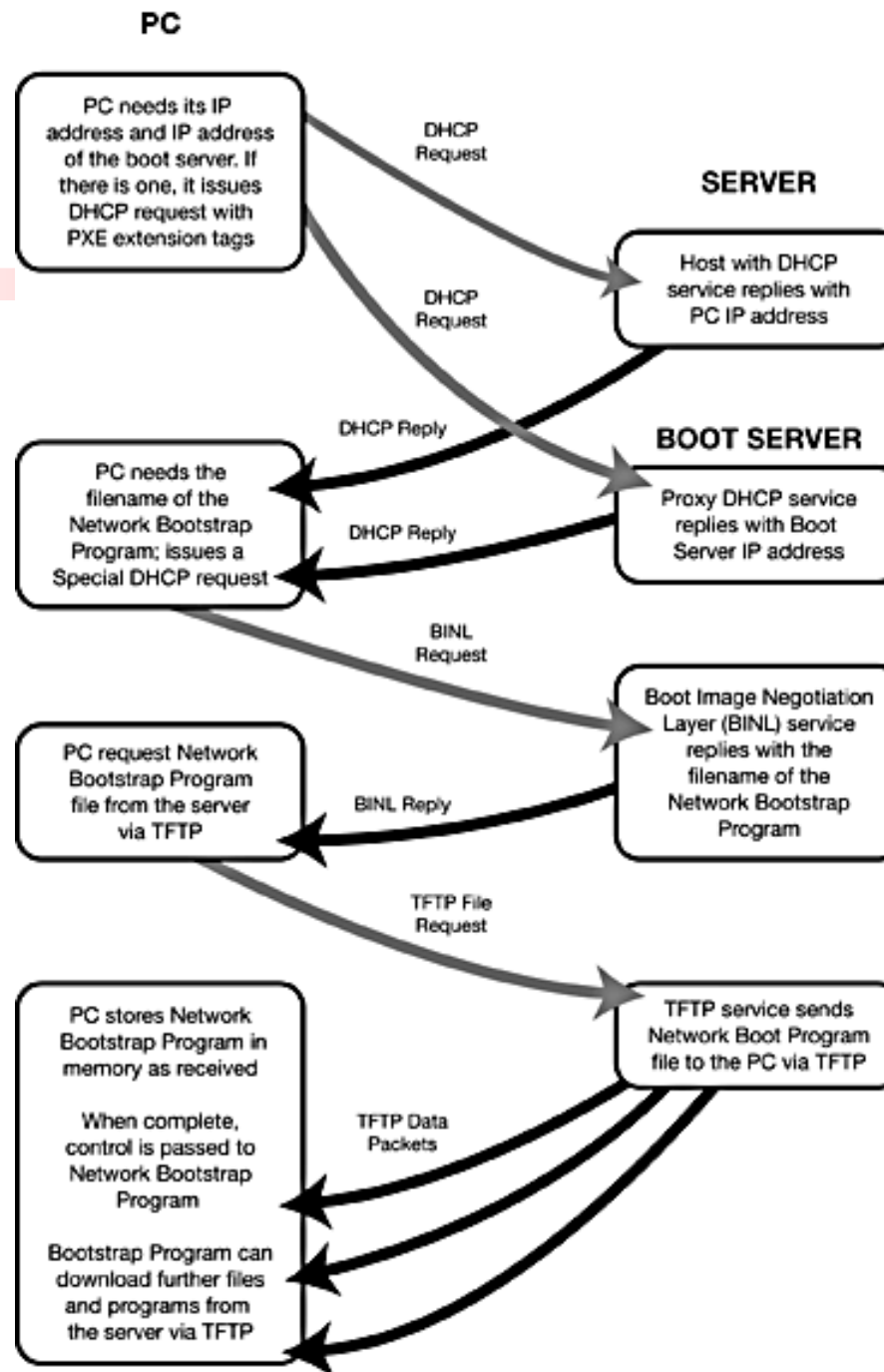
- ◆ PXE: is proprietary specification by Intel to make network booting possible
- ◆ Need a boot server to contain the root file system
- ◆ Boot server can act as a
 - ◆ dhcp server
 - ◆ tftp server
 - ◆ NFS server

How Does PXE Work?

- ◆ Obtain IP address from DHCP server
- ◆ Find PXE boot server (or DHCP server)
- ◆ Download network bootstrap program via TFTP
- ◆ Execute NBP (pxelinux)
 - ◆ Initial ram disk
 - ◆ Linux kernel
 - ◆ Root device mounted over NFS

PXE API





Make the Boot Server Ready - TFTP

- ◆ Any tftp server will do if “tsize” option is supported (RFC 1784/RFC 2349)
 - ◆ Use tftp RPM from distribution
 - ◆ Download sources from <http://www.kernel.org/pub/software/tftp>
- ◆ `mkdir /tftpboot`
- ◆ Edit `/etc/inetd.conf` or `/etc/xinetd.conf/tftp`
 - ◆ `Server_args = -s /tftpboot`

Make the boot server ready - DHCP

- ◆ Install DHCP server v3
 - ◆ RPM from you distribution
 - ◆ Internet Software Consortium (www.isc.org)
- ◆ Make a configuration file which allows
 - ◆ booting
 - ◆ bootp
 - ◆ additional PXE extentions

/etc/dhcp.conf

- ◆ `allow booting;`
- ◆ `allow bootp;`
- ◆ `option domain-name-servers 10.1.1.1;`
- ◆ `option routers 10.1.1.1;`
- ◆ `option dhcp-max-message-size 2048;`
- ◆ `option tftp-server-name "10.1.1.2";`
- ◆ `next-server 10.1.1.2;`
- ◆ `server-name "10.1.1.2";`
- ◆ `filename "/pxes/pxelinux.0";`
- ◆ `option root-path "/tftpboot/_mkcdrec";`

Make the Boot Server Ready - NFS

- ◆ NFS server software needed – install from distro
- ◆ Enable the nfs-server service to start at boot time
 - ◆ On RedHat: `chkconfig --list | grep nfs`
 - ◆ To enable: `chkconfig --level 5 nfs on`
- ◆ `Ps ax`
- ◆ `Rpcinfo -p`

Make the Boot Server Ready – NFS (cnt'd)

- ◆ Create /etc/exports file
 - ◆ `/tftpboot * (rw,no_root_squash)`
- ◆ `/usr/sbin/exportfs -a`
- ◆ `/usr/sbin/showmount -e`
- ◆ Check /etc/hosts.allow (tcp wrappers)
`ALL:10.1.1.0/255.255.255.0`

Install PXELINUX on Boot Server

- ◆ PXELINUX is a bootloader using the PXE network booting protocol
- ◆ Install syslinux package on boot server
 - ◆ Contains syslinux, isolinux, memdisk and pxelinux
 - ◆ Documentation under /usr/share/doc/syslinux*/
- ◆ `mkdir -p /tftpboot/pxes/pxelinux.cfg`
- ◆ `cp /usr/lib/syslinux/pxelinux.0 /tftpboot/pxes`
- ◆ PXE config file in /tftpboot/pxes/pxelinux.cfg/

PXE Configuration File

- ◆ Configuration file lives in /tftpboot/pxes/pxelinux.cfg/
- ◆ Name is MAC address of client PC NIC, or
- ◆ Part of the MAC address of client PC NIC
 - ◆ gethostip 10.1.1.209
 - ◆ E.g. 10.1.1.209: 0A0101D1, 0A0101D, 0A0101, 0A010, 0A01, 0A0, 0A, 0, default
- ◆ Generic name is “default”
- ◆ Content is very similar to syslinux syntax

Client PC: install mkCDrec

- ◆ Purpose is
 - ◆ to make backups on boot server
 - ◆ create bootable root file system on boot server
- ◆ client PC with PXE (F12 key) capabilities
- ◆ Requirements
 - ◆ Linux kernel ability to use /dev/nfs as root
CONFIG_ROOT_NFS=y
 - ◆ mkCDrec v0.7.6 or higher

Edit mkCDrec's Config.sh

- ◆ ISOFS_DIR=/mnt/_mkcdrec/backup
- ◆ Check KERNEL_APPEND
 - ◆ Mandrake needs “ devfs=mount”
- ◆ DEVFS=1 (when devFS is used)
- ◆ BURNCDR=n (do not burn a CD-R)
- ◆ FORCE_DHCP_SUPPORT=N
 - ◆ PXE protocol will get a IP address from the dhcp server anyway

NFS mount the backup directory

- ◆ On the boot server (BS):

- ◆ `mkdir /tftpboot/_mkcdrec`

- ◆ On Client PC:

- ◆ `mount BS:/tftpboot /mnt`

- ◆ `mkdir /mnt/_mkcdrec/backup`

- ◆ Must succeed, otherwise root has permission problems

Run mkCDrec with the backup option

- ◆ `cd $MKCDREC_DIR` (e.g. `/var/opt/mkcdrec`)
- ◆ `make path DESTINATION_PATH=/mnt/_mkcdrec/backup`
- ◆ `make`

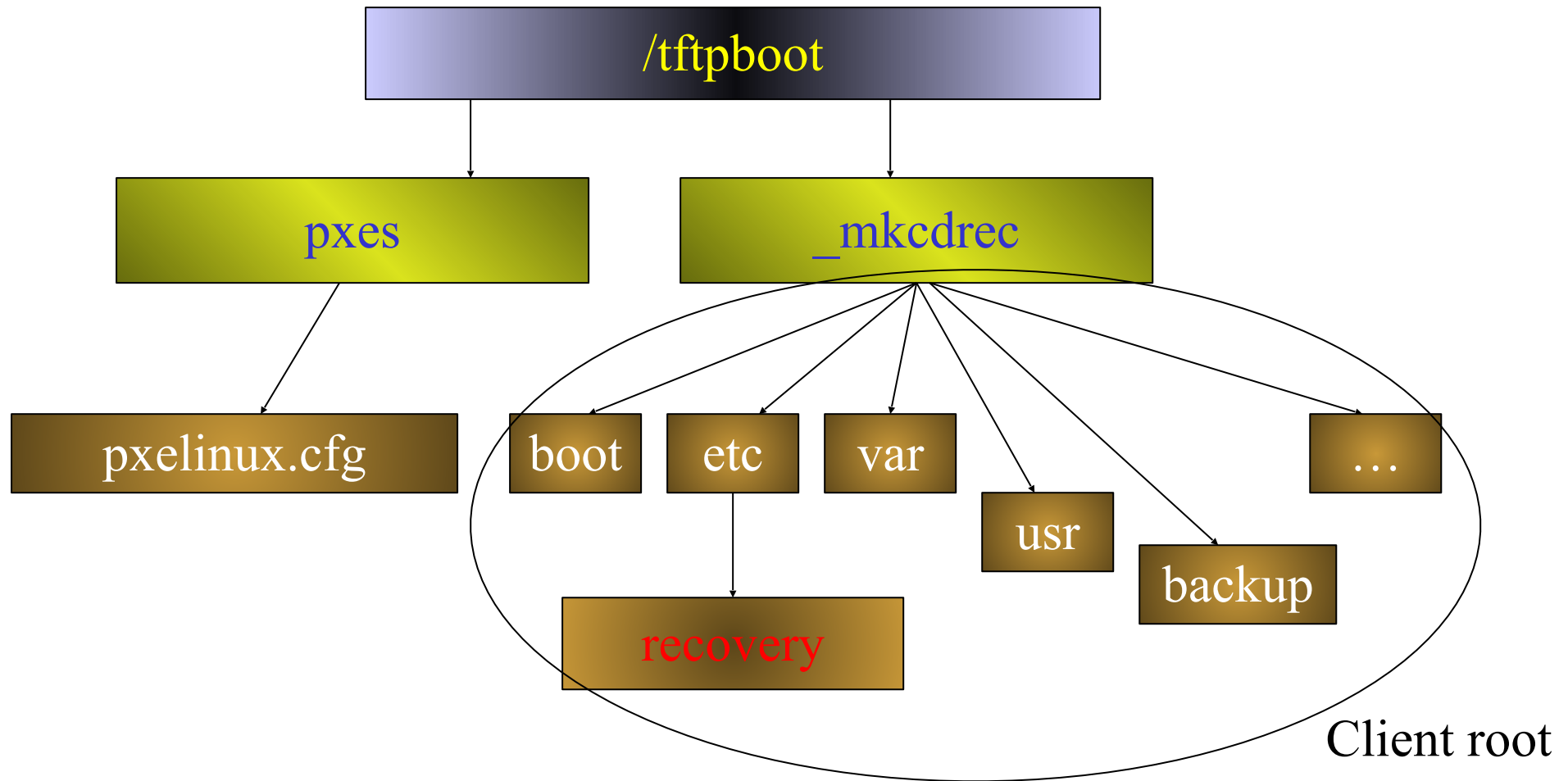
```
mkCDrec v0.7.5 - Backing up your partitions
```

```
Enter the destination path, e.g. /foo  
/mnt/_mkcdrec/backup
```

After successful run of mkcdrec

- ◆ Do **not** execute a “make clean” !!
 - ◆ Would remove content of /mnt/_mkcdrec/backup
- ◆ cd into mkcdrec’s contributions directory
- ◆ Execute `./prepPXE.sh /mnt/_mkcdrec/backup`
 - ◆ Script sets up the PXE root environment
 - ◆ And the initrd boot part for PXE booting
 - ◆ Moves linux kernel to boot directory
 - ◆ Creates pxelinux default file

Layout of /tftpboot on boot server



Client PC – PXE booting

- ◆ At start up press “F12” function key (check BIOS to enable)
- ◆ DHCP v3.x is PXE aware (no need for a separate PXE daemon)
 - ◆ Host=10.1.1.250 (IP given by DHCP server)
 - ◆ Bootserver=10.1.1.2 (PXE/TFTP bootserver)
 - ◆ Rootserver=10.1.1.2 (NFS server where / lives)
 - ◆ Rootpath=/tftpboot/_mkcdrec (the ROOT file system)

Client PC - booted

- ◆ # is waiting on your commands
- ◆ Like booted from CD
 - ◆ Except for the mkcdrec utilities (still linked to /cdrom)
- ◆ /etc/recovery contains the restore scripts
 - ◆ Start-restore.sh
 - ◆ Clone-dsk.sh
 - ◆ Restore-fs.sh

Other PXE related projects

- ◆ Thinix - Linux Thin Client
- ◆ Linbox Backup Server
- ◆ Stresslinux
- ◆ ClusterKnoppix
- ◆ Freevix - media orientated linux distribution
- ◆ Thinstation

References

- ◆ mkCDrec: <http://mkcdrec.ota.be/>
- ◆ PXE documentation: <http://clic.mandrakesoft.com/documentation/pxe/>
- ◆ Pxelinux: <http://syslinux.zytor.com/>
- ◆ DHCP: <http://www.isc.org/products/DHCP/>